Statistical Analysis of Employee Salary Data Using Mean and Variance

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Fundamentals of Data Science • DSA0405

Research Objectives & Scope

- This empirical study evaluates salary distributions across five departments to assess compensation fairness, detect anomalies, and quantify wage dispersion.
- Evaluate salary distributions across five departments to assess compensation fairness, detect anomalies, and quantify wage dispersion using mean and variance.

Population Summary Metrics

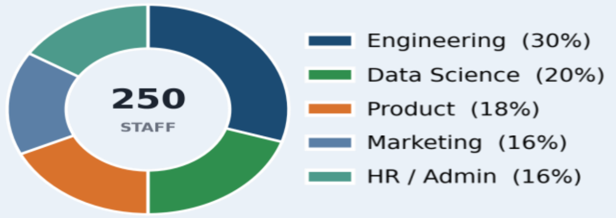
$72,450 Global Mean ($\bar{x}$)

$18,012 Global Std Dev (s)

$324.4M Variance ($s^2$)

24.86% Overall CV

Payroll Budget Distribution



Mathematical Formulation

Sample Mean

$$\bar{x} = (1/n)\, \Sigma\, x_i$$

Sample Variance

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n-1)$$

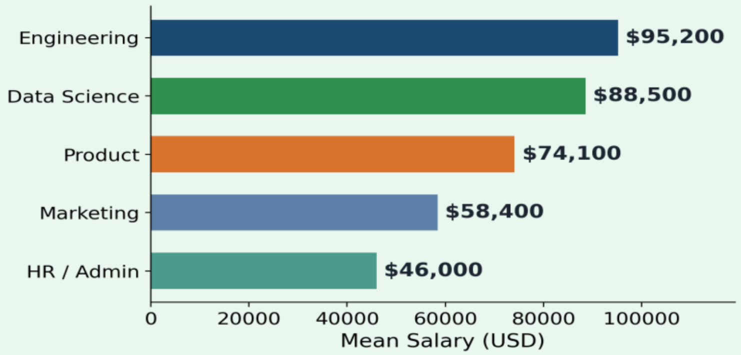
Std. Deviation

$$s = \sqrt{s^2}$$

Coef. of Variation

$$CV = (s / \bar{x}) \times 100$$

Mean Salary by Department



Departmental Statistical Breakdown

Dept	n	Mean	SD	CV
Engineering	75	95,200	19,400	20.4%
Data Sci.	50	88,500	16,200	18.3%
Product	45	74,100	12,500	16.9%
Marketing	40	58,400	9,800	16.8%
HR/Admin	40	46,000	6,200	13.5%
Total/Avg	**250**	**72,450**	**18,012**	**24.8%**

Statistical Interpretations

- Right-skewed: mean ($72.4k) sits well above median ($68.0k) — positive skew.
- Engineering shows extreme variance ($s^2 = 3.76\times10^8$) from a broad seniority ladder.
- ANOVA: $F = 42.18$, $p < 0.001$ — rejects H_0, confirming real pay differences by dept.

Methodology & Assumptions

- Roughly normal distribution, confirmed via Shapiro–Wilk test.
- Independent observations maintained across all 5 departments.
- Compute salary statistics to assess pay equity.

Ingest HR Logs → **Clean** No Outliers → **Stats** Variance → **Equity** CV Check

95% CI & Strategic Actions

95% CI for μ: **[$70,217 – $74,683]**

- Standardize Engineering pay bands (CV 20.4%).
- Lift entry-level HR pay to market floors.
- Run quarterly variance equity audits.

Tech stack: Python · Pandas · SciPy · Excel/Power BI